

Jing Xu (许靖)

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EDUCATION

Tsinghua University <i>Ph.D. in Computer Science at IIIS</i> • Advisor: Andrew Chi-Chih Yao	2021.9 – 2026.7 (expected)
Peking University <i>B.S. in Artificial Intelligence at EECS (Turing Class, with summa cum laude)</i> • Overall GPA: 3.87/4.00 • Ranking: 1/93 • Advisor: Liwei Wang	2017.9 – 2021.7

RESEARCH INTERESTS

My research lies at the intersection of theoretical and applied machine learning. On the theoretical side, I am interested in establishing provable guarantees for the generalization and optimization of machine learning algorithms. On the empirical side, I have hands-on experience with large-scale LLM pre-training and am committed to designing efficient optimization algorithms that improve the scalability and performance pre-training.

I also have in-depth practical experience in quantitative research and have interned at top-tier quantitative trading firms, including Citadel Securities and Jump Trading.

I have worked on topics including:

- Efficient and stable optimizers for LLM pre-training.
- Adaptation of LLMs, e.g., parameter-efficient fine-tuning and scalable model merging.
- Generalization guarantees, implicit bias, and corresponding empirical signals in machine learning.
- Upper and lower convergence bounds for optimization algorithms on structured problems.

INDUSTRY EXPERIENCE

Citadel Securities <i>Quantitative Research Intern</i> • Built LLM pipelines to extract signals and build alphas from text-based alternative dataset. Received return offer.	2025.6 – 2025.9
Moonshot AI <i>Machine Learning Intern at pre-training team</i> • Participated in developing efficient and stable optimization algorithm (e.g. MuonClip and other variants of Muon) for LLM. • Got hands-on experience with LLM pre-training, covering aspects including scaling, architecture, data.	2025.2 – 2025.6
Jump Trading <i>Quantitative Research Intern</i>	2024.6 – 2024.8

- Conducted alpha analysis for China’s stock market.
- Developed a volatility prediction model.
- Designed and implemented a transaction cost prediction model.

PUBLICATIONS & TECHNICAL REPORTS

(* denotes equal contribution)

1. **Kimi k2: Open agentic intelligence**
Kimi Team *et al.*
2. **NeuralDB: Scaling Knowledge Editing in LLMs to 100,000 Facts with Neural KV Database**
Weizhi Fei, Hao Shi, **Jing Xu**, Jingchen Peng, Jiazheng Li, Jingzhao Zhang, Bo Bai, Wei Han, Zhenyuan Chen, Xueyan Niu
The Fourteenth International Conference on Learning Representations (ICLR 2026)
3. **Scalable Model Merging with Progressive Layer-wise Distillation**
Jing Xu, Jiazheng Li, Jingzhao Zhang
Forty-Second International Conference on Machine Learning (ICML 2025)
4. **Understanding Nonlinear Implicit Bias via Region Counts in Input Space**
Jingwei Li*, **Jing Xu***, Zifan Wang, Huishuai Zhang, Jingzhao Zhang
Forty-Second International Conference on Machine Learning (ICML 2025)
5. **Near-Optimal Methods for Convex Simple Bilevel Problems**
Huaqing Zhang*, Lesi Chen*, **Jing Xu**, Jingzhao Zhang
The Thirty-ninth Annual Conference on Neural Information Processing Systems (Neurips 2024)
6. **Random Masking Finds Winning Tickets for Parameter Efficient Fine-tuning**
Jing Xu, JingZhao Zhang
The Forty-first International Conference on Machine Learning (ICML 2024)
7. **On Bilevel Optimization without Lower-level Strong Convexity**
Lesi Chen*, **Jing Xu***, JingZhao Zhang
The Thirty-seventh Annual Conference on Learning Theory (COLT 2024)
8. **Towards Data-Algorithm Dependent Generalization Analysis: a Case Study on Overparameterized Linear Regression**
Jing Xu*, Jiaye Teng*, Yang Yuan, Andrew C Yao
The Thirty-eighth Annual Conference on Neural Information Processing Systems (Neurips 2023)
9. **Quantifying the Variability Collapse of Neural Networks**
Jing Xu*, Haoxiong Liu*
The Fortieth International Conference on Machine Learning (ICML 2023)
10. **Faster Gradient-Free Algorithms for Nonsmooth Nonconvex Stochastic Optimization**
Lesi Chen, **Jing Xu**, Luo Luo
The Fortieth International Conference on Machine Learning (ICML 2023)

HONERS & AWARDS

- Ubiquant Scholarship (2024.9)
- IIIS Scholarship (2022.9 & 2023.9)
- Toyota Scholarship (2023.9)
- Excellent Graduate of PKU (2021.7)
- John Hopcroft Scholarship (2020.9)
- Turing Class Scholarship (2019.9)
- Award for Academic Excellence at PKU (2018.9)
- May 4th Scholarship at PKU (2018.9)

SKILLS

- **English Proficiency:** TOEFL iBT: 107(Reading: 30, Listening: 29, Speaking: 24, Writing: 24), GRE: 332
- **Coding:** I am familiar with modern machine learning frameworks such as PyTorch. I have hands-on experiences of customizing distributed training frameworks such as Megatron.

TEACHING ASSISTANT EXPERIENCES

1. Mathematics for Computer Science
Taught by Professor Andrew Chi-Chih Yao, Tsinghua University, 2022~2023 Spring
2. Introduction to Optimization
Taught by Professor JingZhao Zhang, Tsinghua University, 2022~2023 Autumn
3. Introduction to Computer Systems
Taught by Professor Chenren Xu, Peking University, 2019~2020 Autumn

SERVICES

- Served as a reviewer of ICML2022, 2024, 2025, Neurips2023, 2024, 2025, ICLR2024, 2025, CVPR2024, AAAI2025, AISTATS 2025